ACUSTRA / QUICKSTART

Acoustic Guitar Rack Extension - version 1.3.3f2 - Reason 14 or later

Six physically modelled steel strings, a shared bridge and a resonant body. This guide is for the current Acustra Rack Extension and its factory bank.



Make your first sound

1. Create Acustra Acoustic Guitar in the Reason rack and select its track.

2. Load the factory patch Dreadnought Steel.

3. Keep the automatic main L/R audio routing, then play a note between MIDI 40 and 84 in Standard tuning.

4. Under Finger, play three/four reachable notes together to hear coordinated plucks. Select Pick for swept pairs and larger strums.

5. Try Picking: Finger, Pick or Thumb. Then turn Touch, Pluck Position or Mute while a phrase plays.

Factory starting point: Dreadnought, Spruce, Standard tuning and Finger. The bank uses 2x oversampling, 10% Room, no Detune and normal piezo-blend polarity. Panel Sustain starts Off.

PLAY AND SHAPE

Construction, touch and independent sustain holds

Guitar / Wood	Parlor, Auditorium, Dreadnought or Jumbo; Spruce or Mahogany. These are variations of one shared physical body.
Tuning	Standard, Drop D, DADGAD or Open G. Each tuning defines the lowest open string through fret 20 of the highest string.
Picking / Touch	Finger, Pick or Thumb changes the plucking contact. Touch moves from softer, darker attacks toward firmer, brighter articulation.
Pluck Position / Age	Move the pluck between bridge and neck; age the strings from fresh and bright toward worn and darker.
Mute / Release Noise	Mute presses the picking hand at the bridge. MIDI CC2 breath control can press harder than the knob. Release Noise adds damping contacts and subtle wound-string brushes.
Wheels / Bend Range	The mod wheel adds tension vibrato. Pitch bend uses the selected 1-12 semitone range; the default is 2 semitones.

Keyswitches

The printed octave names follow the Acustra panel. Use the MIDI note numbers when your keyboard or piano-roll octave labels differ. These notes control playing; they do not sound.

C0 - MIDI 24	Latch downstrokes.
C#0 - MIDI 25	Latch upstrokes.
D0 - MIDI 26	Return to automatic stroke direction.
D#0 - MIDI 27	Toggle an independent sustain hold.

Sustain and release

The front Sustain switch, CC64 sustain pedal and D#0 latch hold independently. Released strings stay held while any source is active. The front Sustain status shows their combined state. Clear every hold source to let held strings damp; Reset Audio clears the keyswitch latch.

Ordinary unsustained key-ups allow a tempo-aware 1/32-note window before damping. A same-pitch re-pick within that window can join the phrase. Releasing the final sustain source starts damping directly. Fretted releases fall in pitch with the lifting fingertip; open strings do not.

Automatic strokes alternate within a dotted eighth at the song tempo, capped at two seconds. Pick single notes alternate too. Finger three/four-string chords pluck together; a latched direction makes them sweep. Smaller swept spans cross each string more slowly.

CAPTURE AND ROUTE

Microphones, pickup blend and rear connections



Stereo mic	Main L/R hears the microphone pair. Stereo Width changes its width; Body changes the radiation balance.
Mono mic	Main L/R hears one microphone in mono. Stereo Width has no effect on this capture.
Piezo	Main L/R hears the modelled under-saddle pickup and preamp. Piezo Mix adds nothing when this capture is selected.

Room adds early reflections and a tail to microphone captures. Piezo Mix blends pickup beneath the microphones. Rear PIEZO BLEND / INVERT reverses only that blend; its selectable front readout shows the setting. The separate PIEZO jack and Piezo-only Capture retain their polarity.

L (MONO) / R	Main Capture outputs, automatically routed. L alone carries the mono sum.
PIEZO	Independent mono pickup output, regardless of Capture. Connect it by hand for separate processing; Output gain still applies.
NOTE / GATE CV	Monophonic sequencer control. Note CV sets pitch and Gate CV triggers a pluck; gate level sets velocity. The inputs support automatic routing.

QUALITY AND EVERYDAY USE

Factory reference, CPU choices and practical checks

1x / 2x / 4x	Rear Oversampling runs the full instrument at 1x, 2x or 4x the host sample rate. New instances and factory patches select 2x.
When changes apply	A quality change waits until keys and CV are released and the silent tail window completes, or applies after Reset Audio. The front readout shows the requested quality.
Latency / CPU	All outputs declare 49 host samples at every quality setting (about 1.02 ms at 48 kHz), compensated by Reason. Higher quality and session sample rates increase CPU use; try 1x when headroom is limited.

At 2x, native tests retained transient deadline overruns at 88.2, 96 and 192 kHz, largely shared by the earlier engine. Representative 48 kHz tests passed. These measurements do not guarantee Reason scheduling; use 1x when session headroom is limited.

Default Dreadnought Steel patch

Construction / player	Dreadnought; Spruce; Standard; Finger. String Age 15%, Pluck Position 28%, Touch 58%, Mute 0%.
Microphones / mix	Stereo mic; Body 82%, Stereo Width 62%, Piezo Mix 20%, Room 10%; Output 0 dB.
Performance / quality	Release Noise 10%; Bend Range 2 semitones; Sustain Off; Oversampling 2x; Detune 0%; Piezo Blend Invert Off.

Three useful starting moves

For a dry microphone sound, set Room to 0%. For a stage pickup sound, select Capture Piezo or load Piezo Stage Steel. For gentle pitch variation, turn rear Detune slightly upward: it gives each played tone a fixed independent random offset, up to +/-20 cents at 100%. Held notes keep their draw.

If a note is missing, check the chosen tuning and playable range, the selected track and MIDI input, audio routing, Output level and CV cabling. If notes keep ringing, check panel Sustain, the pedal and the D#0 latch. Quality changes can wait for the remaining tail.

Factory bank

Dreadnought Steel; Parlor Steel Fingerstyle; Jumbo Steel Pick; Mahogany Steel Thumb; Piezo Stage Steel; Open G Slide-ready; DADGAD Mono Mic; Palm Muted Steel Pick.

Guide scope and sources

Prepared 14 October 2026 for the Rack build 1.3.3f2. Source: actual Rack declarations, wrapper and eight patches, based on Rack main a952c2a4. The unmodified canonical Instrument main pin is 90d2d8c (standalone 1.5.0 playing model). Panel illustrations are the matching 1.3.3f2 Shop images.

The first-sound steps are user instructions. This QuickStart does not claim an interactive playback, song save/reopen, host CPU or portal acceptance test. Separate build and guarded Recon receipts describe only the checks actually exercised.